Friends, thank you for having me. I am honoured to be here among you, and allowed to voice my thoughts.

• Stop trying to copy the brute-force path
• Fund it on the timescale real science needs
• Make intelligence per watt a European principle
• Treat talent retention as core infrastructure
• Make paradigm diversity an explicit policy goal

Today I will make the case that one of the highest value directions for both the European AI Frontier Strategy, and AI frontier science both meet at the idea that 'structure' is the missing ingredient. Not scale of computation. Structure of computation.

I want to start somewhere that can feel rather unfashionable nowadays. That is. In optimism.

Europe has the talent. Europe has always had the talent. If you visit the rooms where frontier AI is actually being built today, the rooms, not the press releases, you will see European faces. By birth, by training, by the universities that shaped them, roughly a quarter to a third of the people on the frontier are ours. The exact figure is contested. The order of magnitude is not.

We did not lose because we were absent. We lost because we had an abundance of riches that we have systematically failing to spend. The talent is here, or it was here last Tuesday, before somebody offered it a visa. What is missing is the structure around it. The patient architecture that turns a brilliant person into a brilliant team, and a brilliant team into a frontier.

And this, I want to say, is a beautiful problem to have. It means we do not need to invent an innovation talent base, like China. We have one. The job we must learn how to do well is setting up the structure to retaining, cultivating and connecting that talent into frontier ecosystems.

Now, the harder truth. We are not going to win this decade on compute. There is no European hyperscaler waiting in the wings. If our plan is to imitate the American strategy with a smaller wallet, we will lose slowly, and we will lose expensively. Let us not do that.

The interesting frontier, the one the labs themselves are quietly worried about, is no longer brute force. The next multiplier is increasingly not another order of magnitude of GPUs, but a better idea of what to do with the computation you already have.

A beyond-scaling research agenda should, focus on, rather than building ever larger static parameter banks, the next generation of systems is likely to allocate computation dynamically, preserve useful state over time, reason compositionally, model the world causally and multi-modally, learn through richer and more biologically grounded dynamics, and improve in sustained feedback loops. In other words, the frontier is shifting from more scale to better organised intelligence. I believe europe should lean in to the resource contraints we have and to our abundance of innovative and creative talent what kind of lobAI can only be built by people who cannot afford to be wasteful?

Our current predicament makes european frontier AI strategy put in a situation very similar to the one nature had to follow to evolve life. That is why constraint is not merely a weakness. Constraint is often where the next architecture comes from.

And this is not only about software. What comes after today's AI is physical AI: robots that can clean your home, cook your food, care for the elderly, build your cities, and repair the systems around you. But the arc runs inward as well as outward. The same logic that gives you a home robot eventually gives you medical robots, then nanobots capable of microsurgery, precise repair, and interventions that start to look like an artificial immune system. And once you are there, you are not far from prompting matter itself.

That is the scale of what is coming. And if Europe is not a leader in that wave, it will not merely miss a market. It will miss the chance to help shape the operating system of the physical world. This is not only an industrial question, or even a technological one. It is a civilisational one. Europe must help author that future, or accept that its values will be authored for it by the United States and China.